FML2ILV – Federated Machine Learning

Session 5: FL Robustness and Cyber-Security

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Glossary **Book** **GitHub**

Motivation

Attack Taxonomy

Running Example: Scalar FL

Perturbed Gradient Descent

Defences: Filtering and Robust Aggregation

Robustness Against Non-i.i.d. Data

Summary

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Learning Goals

After this session you are aware of

- different levels of control on parts of an FL system
- goals and implementations of cyber attacks on FL
- robust aggregation rules for FL algorithms
- effect of FL network topology on containing attacks

Personalized (Egoistic) FL

From now on: we are node i.

We execute $\mathcal{F}^{(i)}$ — and ask: what can go wrong?

$$w^{(i,t+1)} = \arg\min_{w} \left[L_i(w) + \alpha \sum_{i' \in \mathcal{N}^{(i)}} A_{i,i'} \left\| w - w^{(i',t)} \right\|^2 \right]$$

What Node i Can Control

Two design handles in the local update:

- **Who to collaborate with:** exclude suspicious neighbours or reduce $A_{i,i'}$
- **How to use neighbour info:** choice of $L_i(\cdot)$ and α

less influenced by neighbours	more influenced by (potentially malicious) neighbours
$\alpha = 0$	$\alpha \to \infty$

What Node i Cannot Control

$$w^{(i,t+1)} = \arg\min_{w} \left[L_i(w) + \alpha \sum_{i' \in \mathcal{N}^{(i)}} A_{i,i'} \left\| w - \overbrace{w^{(i',t)}}^{?} \right\|^2 \right]$$

Some $w^{(i',t)}$ might be perturbed (non-i.i.d. data) or **adversarially crafted**.

How to avoid collaborating with malicious neighbours?

FL in the Wild: What Can Go Wrong

- devices communicate over **open, untrusted** networks
- communication link may be
  - **eavesdropped** (privacy attack)
  - **tampered with** (denial-of-service attack)
- a device may be
  - **compromised** (hacked)
  - **faulty** (hardware failure)
  - **adversarial** (deliberate attack)
- any device sending *arbitrary* updates is called **Byzantine**

Robustness as Legal Requirement

EU AI Act, Article 15 — Accuracy, robustness and cybersecurity

High-risk AI systems shall be designed [...] to achieve an appropriate level of accuracy, robustness, and cybersecurity [...] throughout their lifecycle.

European Parliament and Council of the EU, *Regulation (EU) 2024/1689 (Artificial Intelligence Act)*, Official Journal of the European Union, 12 Jul. 2024.

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Taxonomy of Attacks on FL

Objectives:

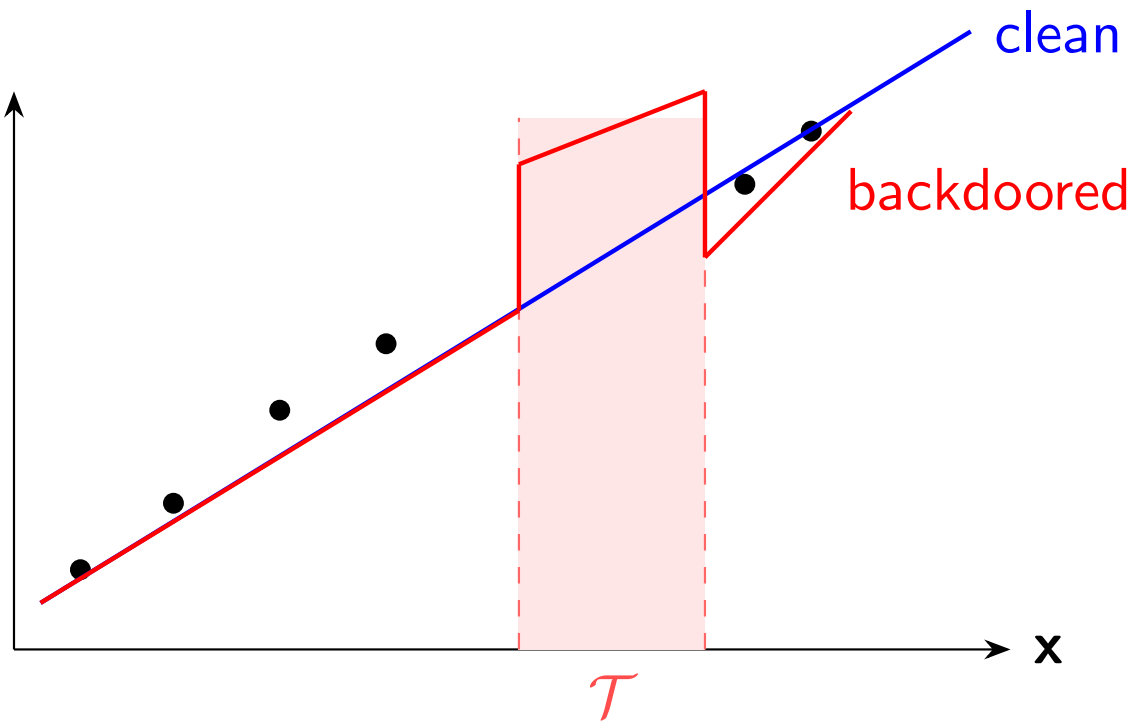
- denial-of-service attack: model performs poorly overall
- backdoor: model behaves well except on trigger pattern $\mathcal{T}$
- privacy attack: adversary learns sensitive info about $\mathcal{D}^{(i)}$

Implementation:

- data poisoning: manipulate data points
- model poisoning: manipulate model parameter updates

Backdoor Attack

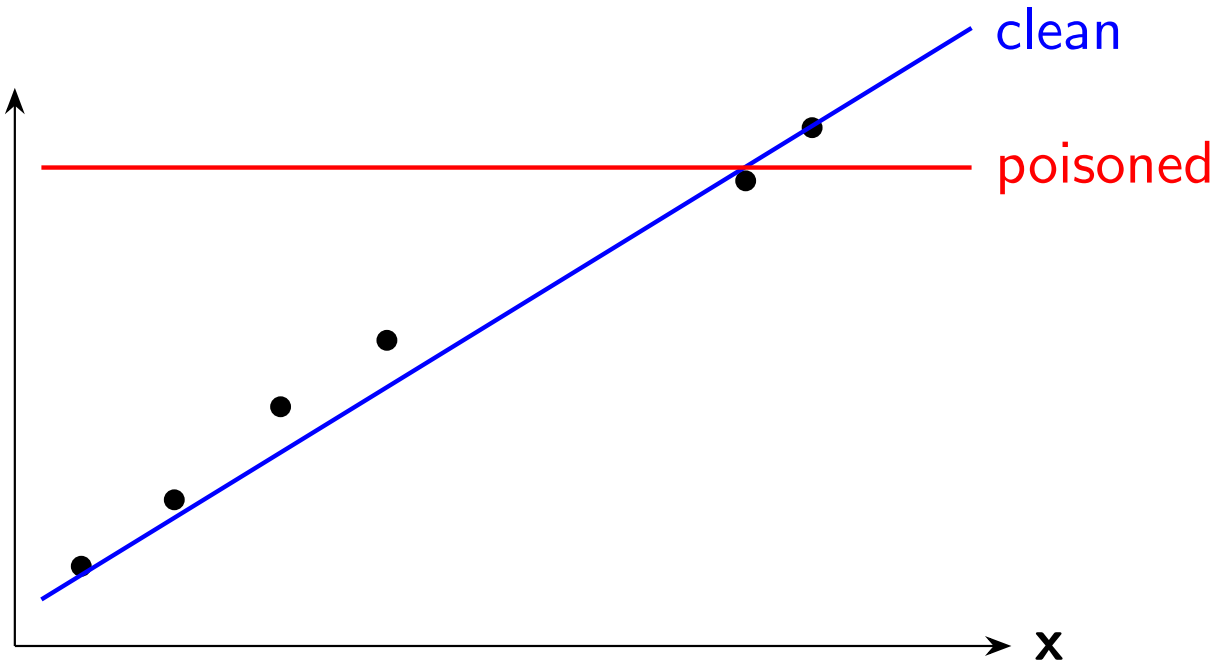
Attacker injects **trigger pattern** $\mathcal{T}$ into trained model:



Normal behaviour outside $\mathcal{T}$; malicious behaviour inside $\mathcal{T}$

Denial-of-Service Attack

Attacker coerces node i into learning params such that model performs poorly **everywhere**:



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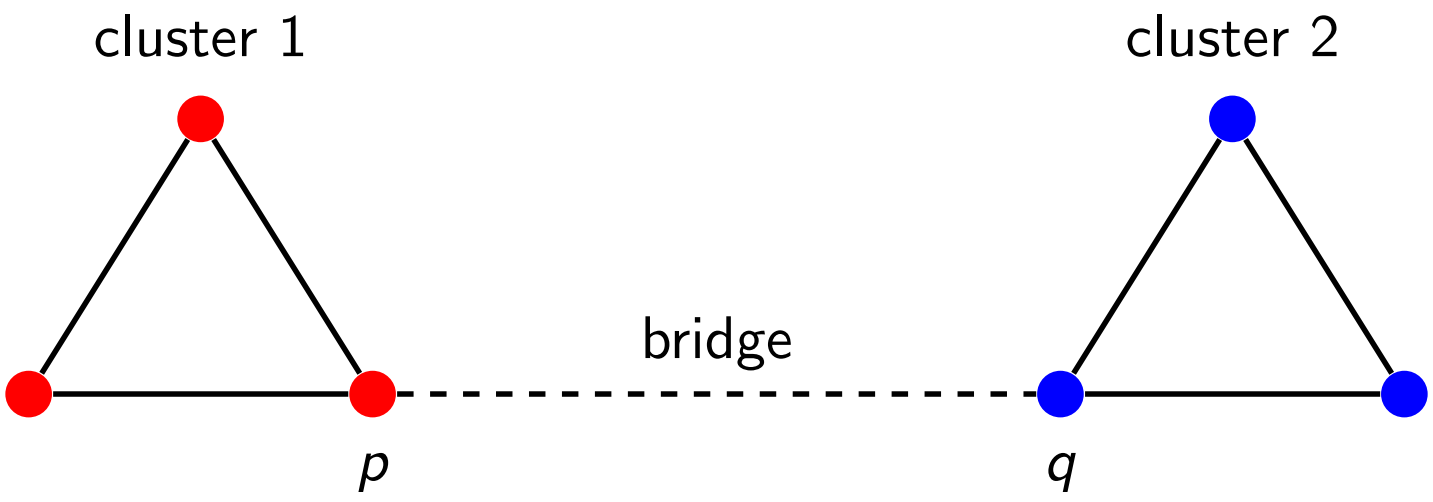
A Simple FL Network



- n devices, $k = 2$ clusters
- local dataset $y^{(i)} = \overline{w}^{(i)} + \mathrm{noise}$
- cluster-wise ground-truth $\overline{w}^{(i)} \in \{\mu^{(1)}, \mu^{(2)}\}$
- local loss $L_i\left(w^{(i)}\right) = \left(y^{(i)} - w^{(i)}\right)^2$

GTVMin: Closed-Form Solution

$$\min_{\mathbf{w} \in \mathbb{R}^n} \sum_{i=1}^{n} (y^{(i)} - w^{(i)})^2 + \alpha \sum_{\{i,i'\} \in \mathcal{E}} A_{i,i'} (w^{(i)} - w^{(i')})^2$$

Zero-gradient $\Rightarrow$ matrix form:

$$(\mathbf{I} + \alpha \, \mathbf{L}) \, \widehat{\mathbf{w}} = \mathbf{y}$$

$\mathbf{I} + \alpha \mathbf{L} \succ 0 \Rightarrow$ unique solution:

$$\boxed{\widehat{\mathbf{w}} = (\mathbf{I} + \alpha \, \mathbf{L})^{-1} \, \mathbf{y}}$$

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Block-wise Gradient at Node i

Gradient of GTVMin w.r.t. $w^{(i)}$:

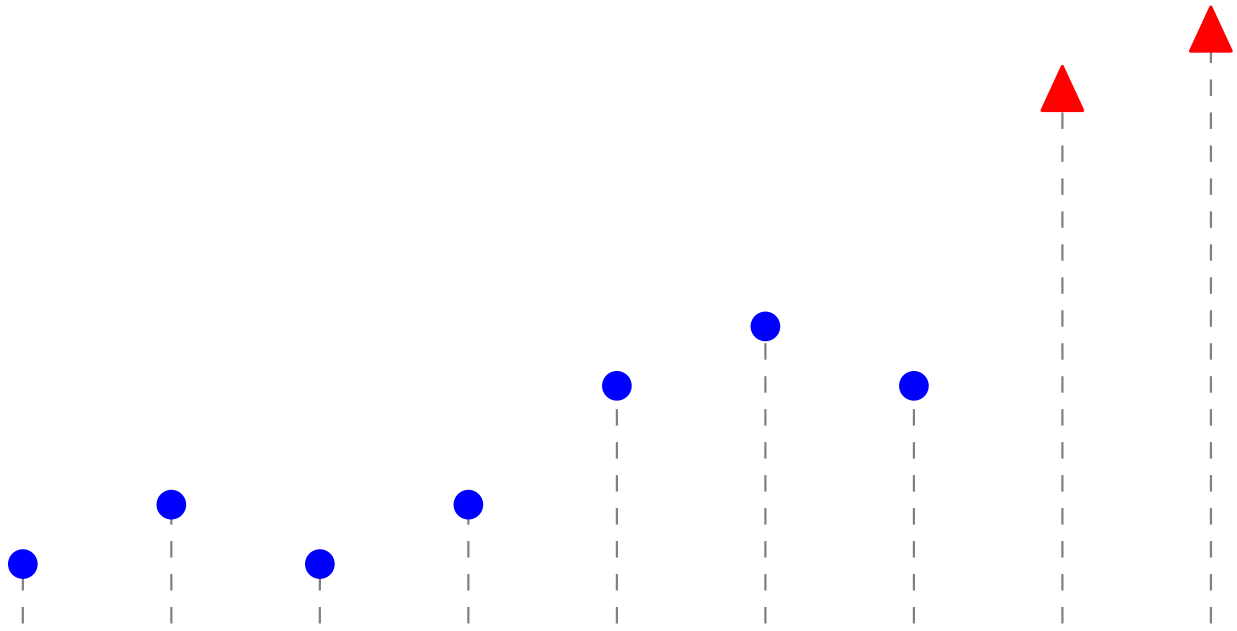
$$g^{(i)}(\mathbf{w}) = \underbrace{\nabla L_i\left(w^{(i)}\right)}_{\text{local gradient}} + \underbrace{2\alpha \sum_{i' \in \mathcal{N}^{(i)}} A_{i,i'}(w^{(i)} - w^{(i')})}_{\text{coupling: needs } w^{(i')} \text{ from neighbours}}$$

Gradient step at round t:

$$w^{(i,t+1)} = w^{(i,t)} - \eta\, g^{(i)}(\mathbf{w}^{(t)})$$

Byzantine Neighbours

Byzantine devices $i' \in \mathcal{A}$ send *arbitrary* $w^{(i',t)}$ to node i:



• honest $i' \in \mathcal{N}^{(i)} \setminus \mathcal{A}$ ▲ Byzantine $i' \in \mathcal{N}^{(i)} \cap \mathcal{A}$

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Two Strategies Against Byzantine Nodes

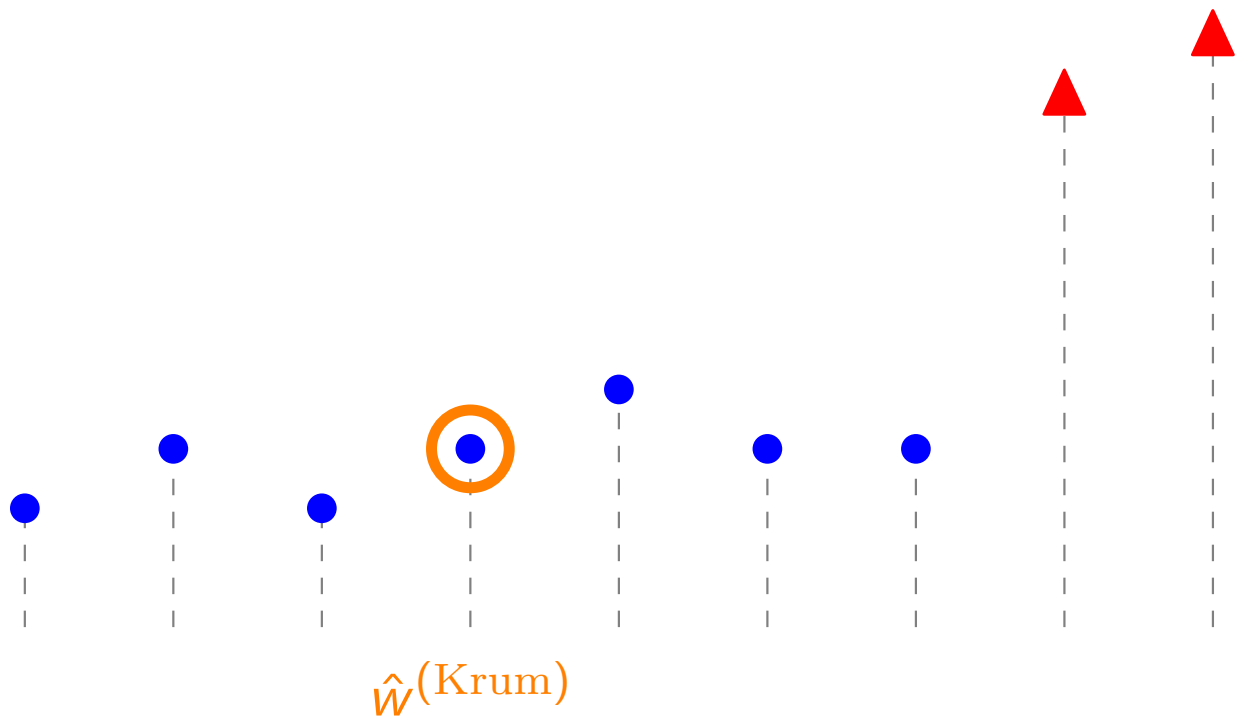
1. **Filter** suspicious neighbour updates
2. **Robust aggregation:** replace sum with median or trimmed mean

Krum

Select update with min distance to nearest neighbours:

$$\hat{w}^{(\mathrm{Krum})} = w^{(i^*)}, \quad i^* = \arg\min_{i' \in \mathcal{N}^{(i)}} \sum_{i'' \in \mathcal{K}^{(i')}} \left(w^{(i',t)} - w^{(i'',t)}\right)^2$$

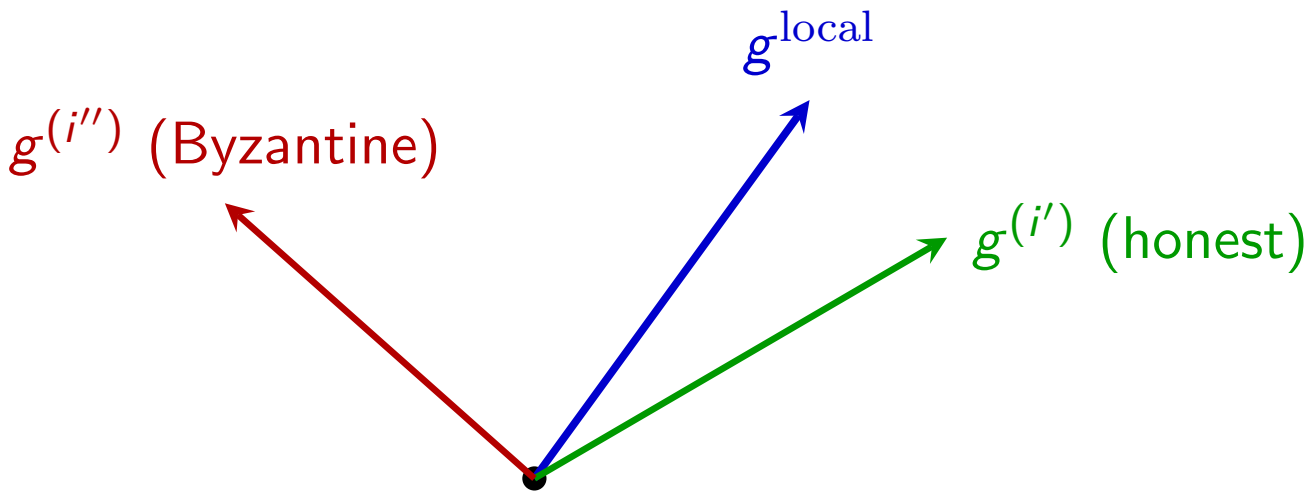
$\mathcal{K}^{(i')}$: set of k nearest neighbours of i'



FLTrust

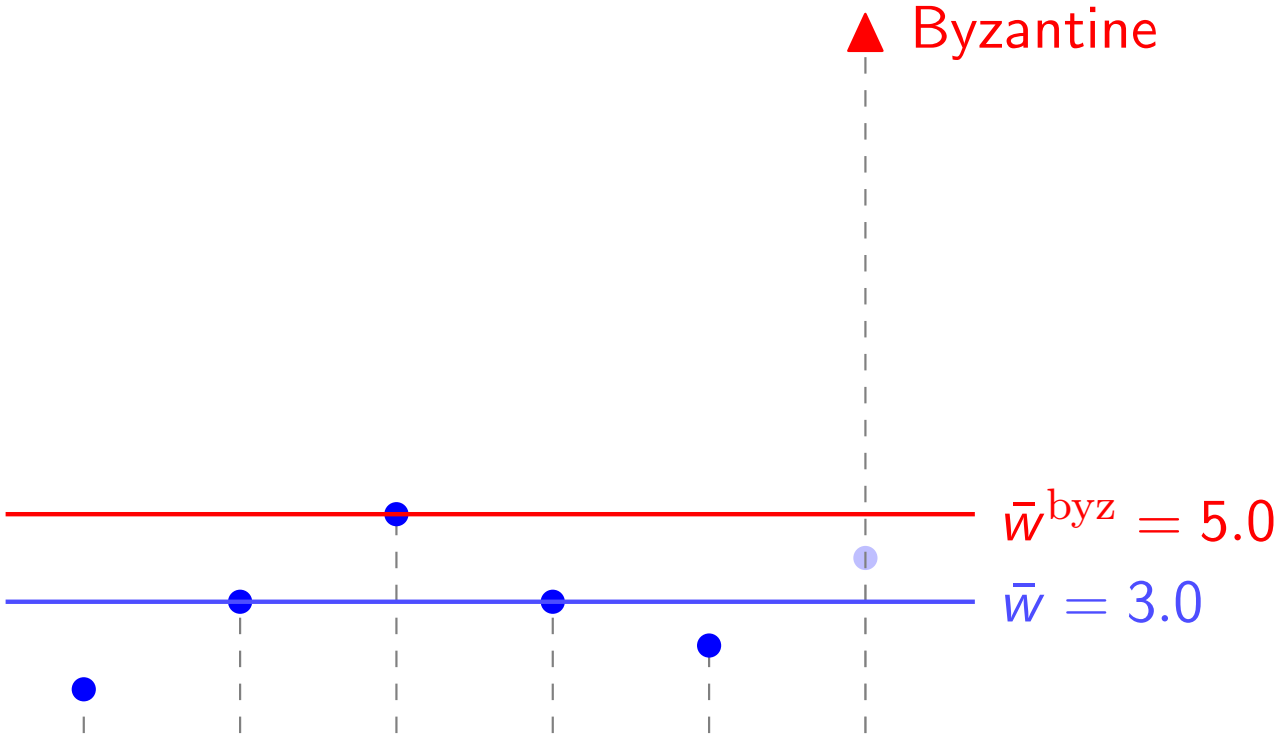
Use own local gradient as **reference**; score neighbours by similarity:

Reference: $g^{\text{local}} = \nabla L_i \left(w^{(i,t)}\right)$
Neighbour contribution: $g^{(i')} = \alpha\, A_{i,i'}(w^{(i,t)} - w^{(i',t)})$
Filter: weight by $\mathrm{ReLU}\big(\cos\langle g^{(i')}, g^{\text{local}}\rangle\big)$



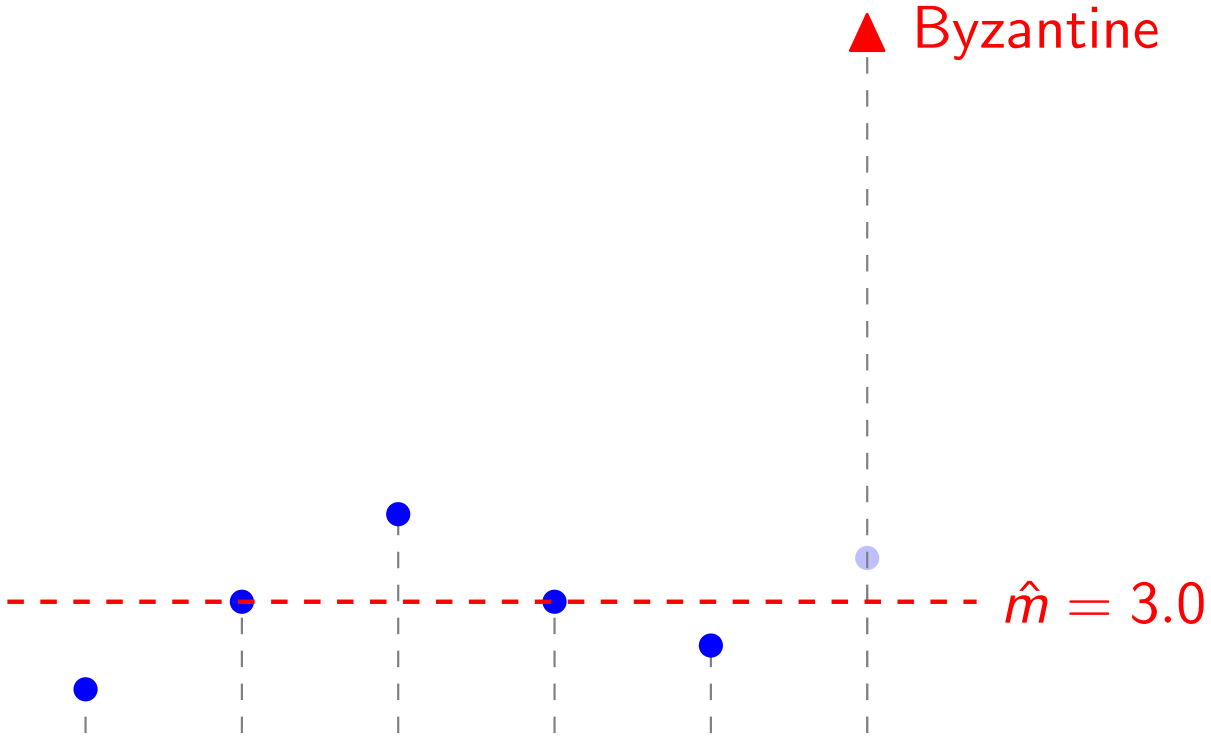
X. Cao et al., *FLTrust: Byzantine-robust FL via Trust Bootstrapping*, NDSS 2021.

Mean is Not Robust



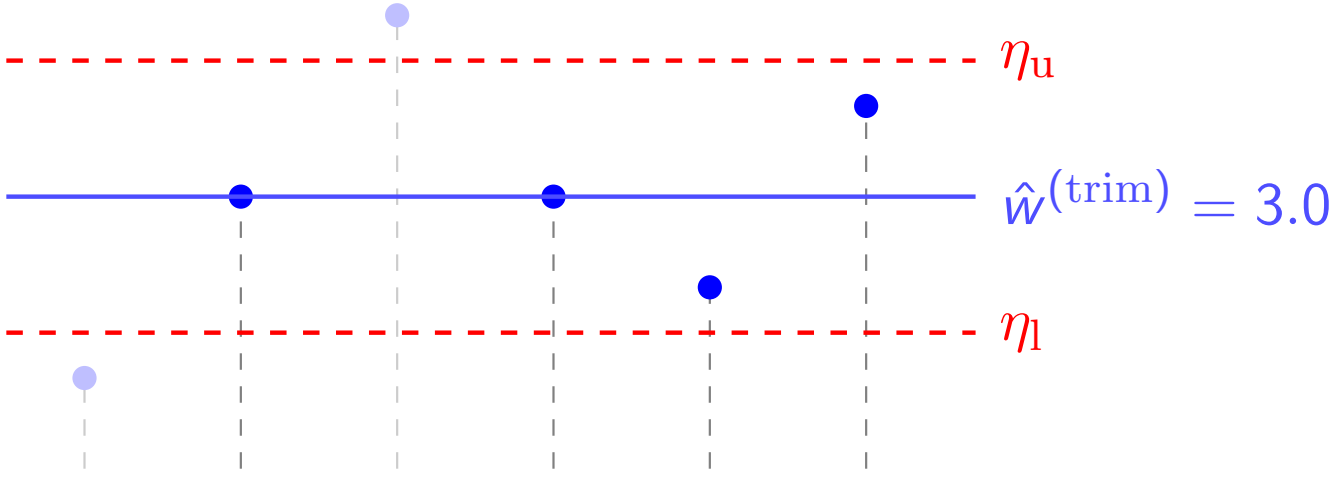
One Byzantine node can shift the mean arbitrarily.

Median



Median unchanged by the Byzantine outlier.

Trimed Mean



Clip values outside $[\eta_{\mathrm{l}}, \eta_{\mathrm{u}}]$, then average.

Robust Aggregation via Gradient Clipping

Replace gradient step with **clipped gradient** + **Huber regularizer**:

$$\mathbf{w}^{(i,t+1)} = \arg\min_{\mathbf{w}} \left[\left\langle \widetilde{\mathbf{g}}^{(t,c)}, \mathbf{w} \right\rangle + R_t\big(\mathbf{w} - \mathbf{w}^{(i,t)}\big) \right]$$

where

$$\widetilde{\mathbf{g}}^{(t,c)} = \frac{\widetilde{\mathbf{g}}^{(t)}}{\|\widetilde{\mathbf{g}}^{(t)}\|} \min\big(h_t, \|\widetilde{\mathbf{g}}^{(t)}\|\big) \qquad \text{(clip to } h_t\text{)}$$

$$R_t(\mathbf{d}) = \begin{cases} \frac{\|\mathbf{d}\|^2}{2\eta} & \|\mathbf{d}\| \le h_t \\ h_t\|\mathbf{d}\| - \frac{h_t^2}{2\eta} & \|\mathbf{d}\| > h_t \end{cases} \qquad \text{(Huber)}$$

J. Zhang & A. Cutkosky, *Unconstrained Robust Online Convex Optimization*, arXiv:2506.12781, 2025.

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From Cyber Security to Statistical Robustness

So far: effects of *malicious* actors (Byzantine).

FL must also be robust against **statistical heterogeneity**: even *honest* nodes may have non-i.i.d. local data, causing updates to diverge.

GTVMin response: personalized model $w^{(i)}$ penalizing only differences with neighbours $\mathcal{N}^{(i)}$.

Effect of Bridge Weight

Split $\mathbf{L} = \mathbf{L}^{(\mathrm{intra})} + \mathbf{L}^{(\mathrm{inter})}$. Using matrix inversion:

$$\widehat{\mathbf{w}} = \underbrace{\mathbf{P}^{(\mathrm{intra})}\,\mathbf{y}}_{\text{cluster-wise solution}} - \underbrace{\alpha\,\mathbf{P}^{(\mathrm{intra})}\,\mathbf{L}^{(\mathrm{inter})}\,\widehat{\mathbf{w}}}_{\text{bridge error}}$$

with $\mathbf{P}^{(\mathrm{intra})} = (\mathbf{I} + \alpha\mathbf{L}^{(\mathrm{intra})})^{-1}$

- $A_{p,q} = 0$: clusters fully decoupled
- $A_{p,q} > 0$: bridge pulls cluster solutions together, error $\propto A_{p,q}$

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- **Attack surface:** neighbour updates can be manipulated by Byzantine nodes
- **Attacks:** backdoor, denial-of-service attack
- **Filtering:** Krum (nearest-neighbour distance)
- **Robust aggregation:** median and trimmed mean instead of mean
- **Non-i.i.d. robustness:** GTVMin solution decomposes into cluster-wise part + bridge error $\propto A_{p,q}$